**Volkan Ozten**
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## Skills:

- Embedded software development with C, C++, Python, C#, MATLAB, and RISC-V Assembly
- Embedded Linux and Windows development environments; Git/GitHub version control and collaborative workflows
- FPGA and firmware development with Verilog/SystemVerilog, Vivado Design Suite, Vitis, and Digilent Nexys A7-100T
- Software/hardware integration, interface debugging, system testing, and technical documentation
- Laboratory testing with oscilloscopes, waveform generators, DMMs, soldering equipment, MATLAB, and LabVIEW

## Experience:

**Consolidated Ocean Technologies, Inc. – Electrical Engineering Intern**
*July 2023 – Present*

- Developed and integrated a C# hardware interface between a remote controller and an AUV's primary control algorithm after evaluating controller hardware and system requirements.
- Diagnosed RS-232 protocol and interface issues between AUV embedded systems and a GPS flasher unit, enabling reliable bidirectional communication.
- Support cross-disciplinary development and laboratory debugging of mechanical, electrical, and software interfaces for a control system housing an inertial navigation system.
- Engineered and documented AUV electrical architecture, including signal-aware cable routing, wire-gauge selection, connector selection, battery packs, and power distribution systems.

**Capstone Design Project – Handheld Thermal Imaging System w/ External Calibration Protocol**
*UC Santa Barbara | Partnered with Teledyne FLIR, September 2024 – June 2025*

- Designed and implemented embedded C++ and Python software integrating thermal-camera input, image-processing algorithms, display output, and user-interface control across two processors.
- Evaluated Broadcom BCM2710A1 and BCM2711 embedded platforms and defined a system architecture balancing size, power, performance, and peripheral-interface requirements.
- Integrated and tested software, electrical, and mechanical subsystems with a multidisciplinary team; performed power characterization, sensor calibration, and thermal validation under load.
- Awarded **Best Multidisciplinary Project** among all UCSB 2025 Engineering Capstone teams.

**Sonatech, LLC – Assistant Test Technician**
*July 2023 – September 2023*

- Executed laboratory acceptance tests for electronic transponders using MATLAB and LabVIEW, verifying signal fidelity, performance, and required product behavior.
- Expanded and optimized MATLAB test procedures to improve repeatability and reliability of transponder signal verification.
- Diagnosed and corrected faults in test procedures, instrumentation, and transponder hardware.

## Education:

**University of California, Santa Barbara**
*M.S. in Computer Engineering – Expected June 2026*
Focus: Embedded Systems, Computer Architecture, and VLSI

*B.S. in Electrical Engineering – Graduated June 2025, GPA of 3.66*
Honors: Dean's List, **Best Multidisciplinary Capstone Project**

## Additional Experience:

- UCSB IEEE Member
- Tutor for AP Physics 1 & 2 in High School